

Assignment 10 :

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Code explanation:

a. Define Images of three:

Here , the three images named in my code image first , image second , image third are being defined using the array by NumPy library . Basically, these images are a second order 2×2 matrix which is represented using the pixel values.

b. Defining the coordinates:

Here , two sets of coordinates have been defined named 'horizontal coordinates newer ' and the 'vertical coordinates new ' are being defined to detect the horizontal and the vertical end points of the image.

c. Definition of the function which does the work :

The function 'apply_coordinates_to_image_new' is being defined in order to calculate the features of an image which is based on the coordinates provided . What it does is basically to perform the element wise multiplication and the summation .

d. Visualization of the image :

Here , the image is then visualized using the pyplot library and the imshow() which is available from matlob . the role is display the image with the pixel values as colors.

e. Calculation the feature and display:

Here ,the function 'apply_coordinates_to_image' is being called twice which is done to calculate the horizontal and the vertical end points of the features scores and then printed.

So, this code overall code defines how to define the images , and then specify the coordinates for the feature selection and then visualize the images by calculating the features which are based on coordinates.

The execution code :

```
import numpy as num
from matplotlib import pyplot as plt

# New images
imageee_first = num.array([num.array([120, 120]), num.array([90, 90])])
imageee_second = num.array([num.array([120, 120]), num.array([60, 0])])
imageee_third = num.array([num.array([120, 60]), num.array([120, 0])])

# New cooredinates for detecting horizontal and vertical end points
horizontal_coordinates_new = num.array([num.array([4, 4]), num.array([-4, -4])])
vertical_coordinates_new = num.array([num.array([4, -4]), num.array([4, -4])])

# Function to apply the set points to image
def apply_the_coordinates_to_image_new(image_new, coordinates_new):
    return num.sum(num.multiply(image_new, coordinates_new))

# plotting the image .
plt.imshow(imageee_first)
plt.axis('off')
plt.title('Image of the First')
plt.show()

# performing a check for the horiz and vert features applied to first image .
print('--- Imagee 1 ---')
print('Horizontalss      extemity      pointss      featurees      scoress:',
      apply_the_coordinates_to_image_new(imageee_first,
      horizontal_coordinates_new))
print('Verticals      extemity      points      features      scoress:',
      apply_the_coordinates_to_image_new(imageee_first,
      vertical_coordinates_new))
print()
```

```

# representing the image 2
plt.imshow(imageeee_second)
plt.axis('off')
plt.title('Image_of the Second')
plt.show()

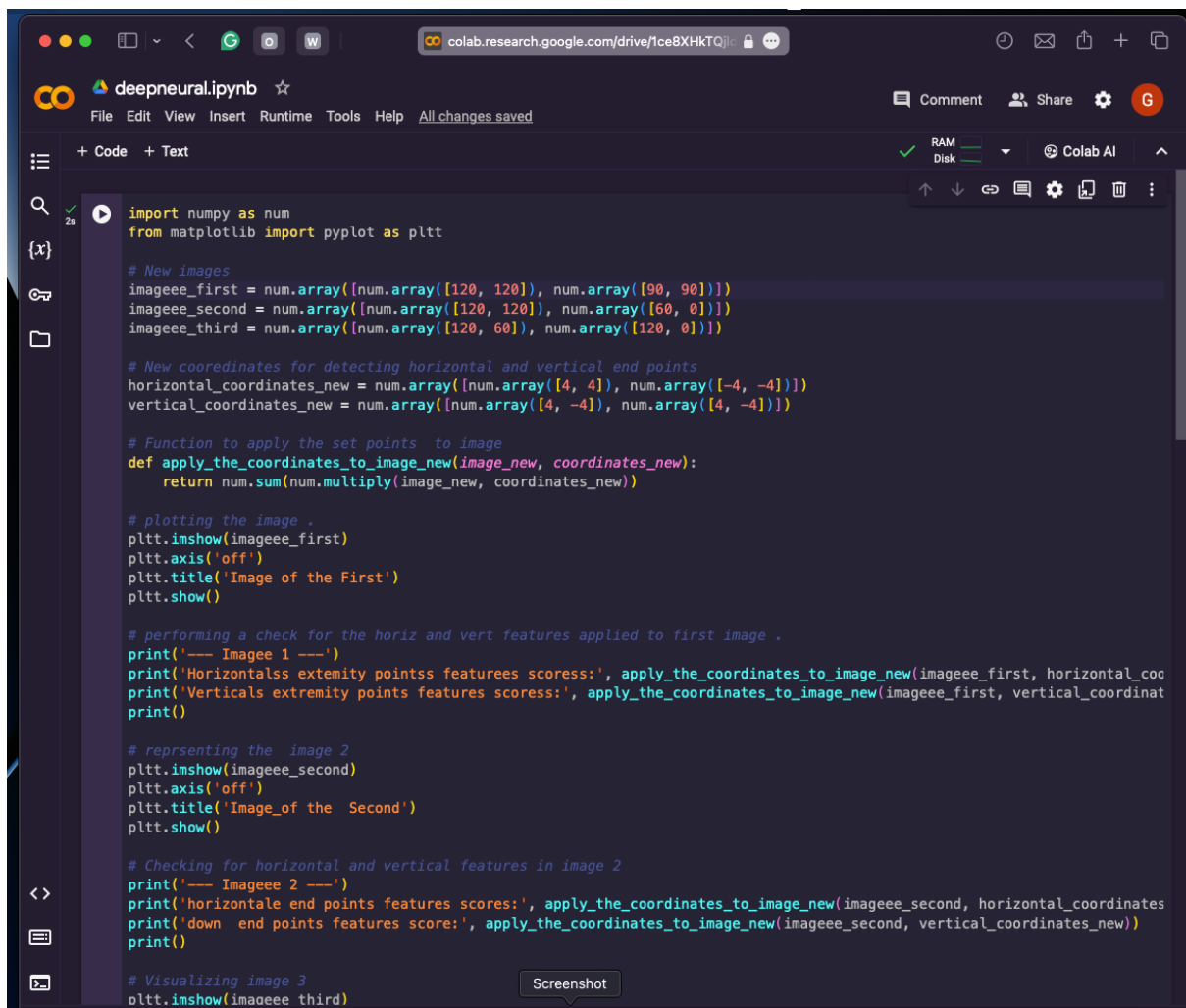
# Checking for horizontal and vertical features in image 2
print('--- Imageee 2 ---')
print('horizontal      end      points      features      scores:',
      apply_the_coordinates_to_image_new(imageeee_second,
      horizontal_coordinates_new))
print('down            end      points      features      score:',
      apply_the_coordinates_to_image_new(imageeee_second,
      vertical_coordinates_new))
print()

# Visualizing image 3
plt.imshow(imageeee_third)
plt.axis('off')
plt.title('Image Third')
plt.show()

# New coordinates for image 3
horizontal_coordinates_new_image3 = num.array([num.array([3, 3]),
num.array([-3, -3])])
vertical_coordinates_new_image3 = num.array([num.array([3, -3]),
num.array([3, -3])])

# Checking for accros and the down features present in the image 3rd
print('--- Image 3 ---')
print('across      end      points      features      score:',
      apply_the_coordinates_to_image_new(imageeee_third,
      horizontal_coordinates_new_image3))
print('down        end      points      features      score:',
      apply_the_coordinates_to_image_new(imageeee_third,
      vertical_coordinates_new_image3))

```



The image shows a Google Colab notebook titled 'deepneural.ipynb'. The interface includes a top navigation bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help' menus. Below the menu is a toolbar with icons for code, text, and other functions. The main area contains a code cell with the following Python code:

```
import numpy as num
from matplotlib import pyplot as plt

# New images
imageee_first = num.array([num.array([120, 120]), num.array([90, 90])])
imageee_second = num.array([num.array([120, 120]), num.array([60, 0])])
imageee_third = num.array([num.array([120, 60]), num.array([120, 0])])

# New coordinates for detecting horizontal and vertical end points
horizontal_coordinates_new = num.array([num.array([4, 4]), num.array([-4, -4])])
vertical_coordinates_new = num.array([num.array([4, -4]), num.array([4, -4])])

# Function to apply the set points to image
def apply_the_coordinates_to_image_new(image_new, coordinates_new):
    return num.sum(num.multiply(image_new, coordinates_new))

# plotting the image .
plt.imshow(imageee_first)
plt.axis('off')
plt.title('Image of the First')
plt.show()

# performing a check for the horiz and vert features applied to first image .
print('--- Imagee 1 ---')
print('Horizontalss extemity pointss featurees scoress:', apply_the_coordinates_to_image_new(imageee_first, horizontal_coo
print('Verticals extemity points features scoress:', apply_the_coordinates_to_image_new(imageee_first, vertical_coordinat
print()

# representing the image 2
plt.imshow(imageee_second)
plt.axis('off')
plt.title('Image_of the Second')
plt.show()

# Checking for horizontal and vertical features in image 2
print('--- Imageee 2 ---')
print('horizontale end points features scores:', apply_the_coordinates_to_image_new(imageee_second, horizontal_coordinates
print('down end points features score:', apply_the_coordinates_to_image_new(imageee_second, vertical_coordinates_new))
print()

# Visualizing image 3
plt.imshow(imageee_third)
```

At the bottom of the code cell, there is a 'Screenshot' button. The interface also shows a left sidebar with navigation icons and a top right area with 'Comment', 'Share', and 'Colab AI' options.

colab.research.google.com/drive/1ce8XHkTQj...

deepneural.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Comment Share Colab AI

+ Code + Text

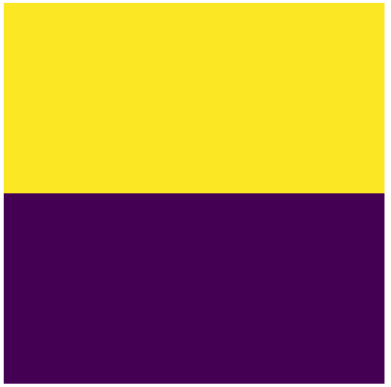
2s

```
# Visualizing image 3
plt.imshow(imageee_third)
plt.axis('off')
plt.title('Image Third')
plt.show()

# New coordinates for image 3
horizontal_coordinates_new_image3 = num.array([num.array([3, 3]), num.array([-3, -3])])
vertical_coordinates_new_image3 = num.array([num.array([3, -3]), num.array([3, -3])])

# Checking for accross and the down features present in the image 3rd
print('--- Image 3 ---')
print('across end points features score:', apply_the_coordinates_to_image_new(imageee_third, horizontal_coordinates_new_in
print('down end points features score:', apply_the_coordinates_to_image_new(imageee_third, vertical_coordinates_new_image3
```

Image of the First



```
--- Imagee 1 ---
Horizontalss extremity pointss featureses scores: 240
Verticals extremity points features scores: 0
```

Image of the Second

Netflix

The screenshot shows a Google Colab notebook interface. At the top, the browser address bar displays 'colab.research.google.com/drive/1ce8XHkTQj...'. The notebook title is 'deepneural.ipynb'. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. The toolbar shows 'Comment', 'Share', and 'Colab AI' options. The notebook content consists of a code cell with the following text:

```
Horizontalss extremity pointss features scores: 240  
Verticals extremity points features scores: 0
```

Below the code cell, there are two images. The first image, titled 'Image of the Second', is a 2x2 grid of colored squares: yellow (top-left), teal (top-right), and purple (bottom-right). The second image, titled 'Image Third', is a 2x2 grid of colored squares: yellow (top-left), teal (top-right), and purple (bottom-right). The output of the code cell shows feature scores for these images:

```
--- Imageee 2 ---  
horizontale end points features scores: 720  
down end points features score: 240
```